Übungsaufgaben: Gleichungen

Löse die Aufgaben in deinem Heft. ★☆☆ einfach · ★★☆ mittel · ★★★ schwer. Die Lösungen stehen auf Seite 2.

Name: _____ Datum: _____

★☆☆ Einfach — Grundwissen

1. $-8x - 12 = -60$
2. $7x - 6 = 1$
3. $5x + 5 = 35$
4. $2x + 5 = -5$
5. $-x + 2 = -5$
6. $3x + 7 = 10$
7. $-2x + 9 = 19$
8. $6x + 4 = -44$
9. $-8x - 11 = 29$
10. $6x + 9 = 51$
11. $8x - 12 = -60$
12. $-2x + 1 = 7$
13. $-5x - 4 = 26$
14. $-6x + 5 = 41$
15. $-7x + 5 = -30$
16. $-9x + 12 = 57$
17. $6x + 10 = -44$
18. $-5x + 7 = -33$
19. $-3x - 4 = 8$
20. $-4x - 1 = -5$
21. $8x + 3 = -37$
22. $4x + 5 = -11$

★★☆ Mittel — Anwenden

23. $8x + 2 = -4x + 62$
24. $-x + 2 = 5x + 14$
25. $4x + 8 = -5x + 62$
26. $-7x + 1 = -x + 31$
27. $-4x + 9 = -5x + 12$
28. $4x - 9 = 3x - 14$
29. $7x - 1 = 41$
30. $2x = -4x + 12$
31. $2x + 6 = -x + 3$
32. $-2x - 6 = x - 27$
33. $7x + 3 = 24$
34. $4x = 2x + 14$
35. $-6x - 7 = -5x + 1$
36. $5x - 7 = 3x - 3$
37. $8x + 8 = -2x + 88$
38. $4x + 4 = 2x - 10$
39. $5x = 4x + 2$
40. $4x - 7 = -5x - 52$
41. $4x + 8 = 5x + 4$
42. $3x + 3 = 2x + 5$
43. $-4x + 5 = 3x + 61$
44. $2x - 6 = -4x - 30$

★★★ Schwer — Knobeln und Transfer

45. $4(2x - 5) = 2x - 23$
46. $\frac{x}{3} - 4 = 1$
47. $(x + 3)(x + 3) = (x + 2)(x - 5) + 55$
48. $-3(3x - 3) = -2x + 30$
49. $\frac{x}{2} + 5 = 9$
50. $(x + 4)(x + 3) = (x + 1)(x - 3) + 24$
51. $3(x + 3) = -x + 17$
52. $\frac{x}{5} - 3 = 0$
53. $(x + 3)(x + 3) = (x + 4)(x - 5) + 57$
54. $5(x + 4) = x + 12$
55. $\frac{x}{5} + 6 = 4$
56. $(x + 1)(x - 2) = (x + 1)(x + 1) - 6$
57. $3(3x - 2) = -2x - 50$
58. $\frac{x}{3} - 2 = -5$
59. $(x + 3)(x - 3) = (x + 3)(x + 1) - 36$
60. $6(x + 2) = 3x + 18$
61. $\frac{x}{2} + 2 = 3$
62. $(x + 3)(x + 3) = (x + 2)(x + 1) - 8$
63. $-3(x + 5) = 2x - 40$
64. $\frac{x}{2} - 4 = -10$
65. $(x + 1)(x - 4) = (x + 2)(x + 1) + 18$

Lösungen: Gleichungen

Lösungen · Seite 2

Vergleiche mit deinen Ergebnissen. Die Nummern entsprechen den Aufgaben auf Seite 1.

★☆☆ Einfach — Grundwissen

1. $-8x = -48 \Rightarrow \boldsymbol{x = 6}$

2. $7x = 7 \Rightarrow \boldsymbol{x = 1}$

3. $5x = 30 \Rightarrow \boldsymbol{x = 6}$

4. $2x = -10 \Rightarrow \boldsymbol{x = -5}$

5. $-x = -7 \Rightarrow \boldsymbol{x = 7}$

6. $3x = 3 \Rightarrow \boldsymbol{x = 1}$

7. $-2x = 10 \Rightarrow \boldsymbol{x = -5}$

8. $6x = -48 \Rightarrow \boldsymbol{x = -8}$

9. $-8x = 40 \Rightarrow \boldsymbol{x = -5}$

10. $6x = 42 \Rightarrow \boldsymbol{x = 7}$

11. $8x = -48 \Rightarrow \boldsymbol{x = -6}$

12. $-2x = 6 \Rightarrow \boldsymbol{x = -3}$

13. $-5x = 30 \Rightarrow \boldsymbol{x = -6}$

14. $-6x = 36 \Rightarrow \boldsymbol{x = -6}$

15. $-7x = -35 \Rightarrow \boldsymbol{x = 5}$

16. $-9x = 45 \Rightarrow \boldsymbol{x = -5}$

17. $6x = -54 \Rightarrow \boldsymbol{x = -9}$

18. $-5x = -40 \Rightarrow \boldsymbol{x = 8}$

19. $-3x = 12 \Rightarrow \boldsymbol{x = -4}$

20. $-4x = -4 \Rightarrow \boldsymbol{x = 1}$

21. $8x = -40 \Rightarrow \boldsymbol{x = -5}$

22. $4x = -16 \Rightarrow \boldsymbol{x = -4}$

★★☆ Mittel — Anwenden

23. $12x = 60 \Rightarrow \boldsymbol{x = 5}$

24. $-6x = 12 \Rightarrow \boldsymbol{x = -2}$

25. $9x = 54 \Rightarrow \boldsymbol{x = 6}$

26. $-6x = 30 \Rightarrow \boldsymbol{x = -5}$

27. $x = 3 \Rightarrow \boldsymbol{x = 3}$

28. $x = -5 \Rightarrow \boldsymbol{x = -5}$

29. $7x = 42 \Rightarrow \boldsymbol{x = 6}$

30. $6x = 12 \Rightarrow \boldsymbol{x = 2}$

31. $3x = -3 \Rightarrow \boldsymbol{x = -1}$

32. $-3x = -21 \Rightarrow \boldsymbol{x = 7}$

33. $7x = 21 \Rightarrow \boldsymbol{x = 3}$

34. $2x = 14 \Rightarrow \boldsymbol{x = 7}$

35. $-x = 8 \Rightarrow \boldsymbol{x = -8}$

36. $2x = 4 \Rightarrow \boldsymbol{x = 2}$

37. $10x = 80 \Rightarrow \boldsymbol{x = 8}$

38. $2x = -14 \Rightarrow \boldsymbol{x = -7}$

39. $x = 2 \Rightarrow \boldsymbol{x = 2}$

40. $9x = -45 \Rightarrow \boldsymbol{x = -5}$

41. $-x = -4 \Rightarrow \boldsymbol{x = 4}$

42. $x = 2 \Rightarrow \boldsymbol{x = 2}$

43. $-7x = 56 \Rightarrow \boldsymbol{x = -8}$

44. $6x = -24 \Rightarrow \boldsymbol{x = -4}$

★★★ Schwer — Knobeln und Transfer

45. $8x - 20 = 2x - 23 \Rightarrow \boldsymbol{x = -\frac{1}{2}}$

46. $\frac{x}{3} = 5 \Rightarrow \boldsymbol{x = 15}$

47. x^2 fällt weg: $6x + 9 = -3x + 45 \Rightarrow \boldsymbol{x = 4}$

48. $-9x + 9 = -2x + 30 \Rightarrow \boldsymbol{x = -3}$

49. $\frac{x}{2} = 4 \Rightarrow \boldsymbol{x = 8}$

50. x^2 fällt weg: $7x + 12 = -2x + 21 \Rightarrow \boldsymbol{x = 1}$

51. $3x + 9 = -x + 17 \Rightarrow \boldsymbol{x = 2}$

52. $\frac{x}{5} = 3 \Rightarrow \boldsymbol{x = 15}$

53. x^2 fällt weg: $6x + 9 = -x + 37 \Rightarrow \boldsymbol{x = 4}$

54. $5x + 20 = x + 12 \Rightarrow \boldsymbol{x = -2}$

55. $\frac{x}{5} = -2 \Rightarrow \boldsymbol{x = -10}$

56. x^2 fällt weg: $-x - 2 = 2x - 5 \Rightarrow \boldsymbol{x = 1}$

57. $9x - 6 = -2x - 50 \Rightarrow \boldsymbol{x = -4}$

58. $\frac{x}{3} = -3 \Rightarrow \boldsymbol{x = -9}$

59. x^2 fällt weg: $-9 = 4x - 33 \Rightarrow \boldsymbol{x = 6}$

60. $6x + 12 = 3x + 18 \Rightarrow \boldsymbol{x = 2}$

61. $\frac{x}{2} = 1 \Rightarrow \boldsymbol{x = 2}$

62. x^2 fällt weg: $6x + 9 = 3x - 6 \Rightarrow \boldsymbol{x = -5}$

63. $-3x - 15 = 2x - 40 \Rightarrow \boldsymbol{x = 5}$

64. $\frac{x}{2} = -6 \Rightarrow \boldsymbol{x = -12}$

65. x^2 fällt weg: $-3x - 4 = 3x + 20 \Rightarrow \boldsymbol{x = -4}$